

Task 1: Props (Product Card System)

Create a **Product Card Component** system:

- Create a `Parent` component
- Create a `ProductCard` child component

- Pass the following data using props:
 - Product Name
 - Price
 - Brand
 - InStock (true/false)

Requirements:

- Show “Available” if product is in stock
- Show “Out of Stock” if not in stock
- Display all data inside a styled card

Task 2: useState (Like Counter + Toggle System)

Create a component that includes:

- A counter (Increase / Decrease)
- A toggle button (Show/Hide text)

Requirements:

- Use `useState` for both features
- When toggle is ON → show “Hello React Students”
- When OFF → hide the text

Task 3: useEffect (API Simulation + Title Change)

Create a component that:

- Displays a loading message first: "Loading data..."

- After 3 seconds, show: "Data Loaded Successfully"
 - Change browser tab title to "React App Loaded"
- Requirements:**

- Use `setTimeout` inside `useEffect`
- Use cleanup function if needed

Task 4: useRef (Previous Value Tracker)

Create a component that:

- Has an input field
- Stores current value
- Also shows previous value of input

Requirements:

- Use `useRef` to store previous value 
- Display:
 - Current Value
 - Previous Value

Task 5: useContext (Language Switcher System)

Create a simple language system:

- Default language: English
- Add a button to switch language:
 - English → "Hello User" ◦
 - Sindhi → "Cha Haal aa Wada"

Requirements:

- Use Context API
- Provide language globally to components
- Show text changes instantly when switching language

Task 1: Props (Product Card System)

❖❖ Parent Component

```
import ProductCard from "../ProductCard";

function Parent() {
  return (
    <div>
      <ProductCard
        name="Laptop"
        price={75000} brand="HP"
        inStock={true}
      />
    </div>
  );
}
export default Parent;
```

❖❖ Child Component (ProductCard)

```
function ProductCard(props) {
  return ( <div
    style={{
      border: "1px solid black",
      padding: "15px", width:
        "250px", borderRadius:
        "10px",
    }}
    >
    <h2>{props.name}</h2>
    <p>Price: {props.price}</p>
    <p>Brand: {props.brand}</p>

    <p>
      Status:{" "}
      {props.inStock ? "Available" : "Out of Stock"}
    </p>
  </div>
  );
}
export default ProductCard;
```

✓ Task 2: useState (Counter + Toggle)

```
import { useState } from "react";
function CounterToggle() {
  const [count, setCount] = useState(0);
  const [showText, setShowText] = useState(false);
  return
```

```

(
  <div>
    <h2>Counter: {count}</h2>

    <button onClick={() => setCount(count + 1)}>
Increase
    </button>

    <button onClick={() => setCount(count - 1)}>
Decrease
    </button>

    <hr />

    <button onClick={() => setShowText(!showText)}>
Toggle Text
    </button>

    {showText && <p>Hello React Students</p>}
  </div>
);
}
export default CounterToggle;

```

Task 3: useEffect (Loading + Title Change)

```

import { useState, useEffect } from "react";

function UseEffectTask() {
  const [message, setMessage] = useState("Loading data...");

  useEffect(() => {
    document.title = "React App Loaded";

    const timer = setTimeout(() => {
      setMessage("Data Loaded Successfully");    },
3000);
    return () =>
clearTimeout(timer);
  }, []);
  return
  (
    <div>
      <h2>{message}</h2>
    </div>
  );
}
export default UseEffectTask;

```

✓ Task 4: useRef (Previous Value Tracker) import

```
{ useState, useEffect, useRef } from "react";

function UseRefTask() {
  const [value, setValue] = useState("");
  const prevValue = useRef("");

  useEffect(() => {
    prevValue.current = value;
  }, [value]);
  return ( <div>
    <input
      type="text"
      value={value}
      onChange={(e) => setValue(e.target.value)}    />

    <h3>Current Value: {value}</h3>
    <h3>Previous Value: {prevValue.current}</h3>
  </div>
  );
}
export default UseRefTask;
```

✓ Task 5: useContext (Language Switcher System)

◆◆ Step 1: Context File

```
import { createContext } from "react";

export const LanguageContext = createContext();
```

◆◆ Step 2: Provider (App.js)

```
import { useState } from "react";
import { LanguageContext } from "../LanguageContext"; import
Home from "../Home";

function App() {
  const [language, setLanguage] = useState("en");
```

```

    const toggleLanguage = () => {
      setLanguage((prev) => (prev === "en" ? "sd" : "en"));
    };
    return (
      <LanguageContext.Provider value={{ language, toggleLanguage }}>
      <div style={{ textAlign: "center", marginTop: "20px" }}>    <button
onClick={toggleLanguage}>
      Switch Language
    </button>

    <Home />
  </div>
  </LanguageContext.Provider>
);
}
export default App;

```

◆◆ Step 3: Consumer Component (Home.js)

```

import { useContext } from "react";
import { LanguageContext } from "../LanguageContext";

function Home() {
  const { language } = useContext(LanguageContext);

  return (
    <div style={{ marginTop: "20px" }}>
    <h2>
    {language === "en"
    ? "Hello User"
    : "Cha Haal aa Wada"}
    </h2>
    </div>
  );
}
export default Home;

```